

Q.21 Differentiate between mechanical extraction and solvent extraction with reference to vegetable oil extraction. (CO2)

Q.22 Discuss the properties and uses of triple super phosphate. (CO4)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain the process of manufacturing of sulfuric acid using contact process with the help of flowsheet. Mention any two engineering problems associated with it. (CO4)

Q.24 Write short notes on any two of following.

a) Ammonium nitrate (CO4)

b) Kraft process (CO2)

c) Different types of cement. (CO3)

d) Piping and instrumentation diagram. (CO1)

Q.25 Explain the manufacturing process of Fatty acids, soaps and glycerine with the help of flowsheet. Mention their properties and uses.

No. of Printed Pages : 4

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Roll No.

3rd Sem / Chemical

Subject : Chemical Technology

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Catalyst used in Haber process for ammonia Production is (CO4)

a) Reduced Iron Oxide b) Nickel

c) Vanadium Pentoxide d) Silica gel

Q.2 Which of the following is an Example of fats? (CO2)

a) Glyceryl trioleate b) Vegetable ghee

c) Coconut oil d) Ground nut oil

Q.3 Wine is obtained after alcoholic fermentation of (CO3)

a) Rice b) Wheat

c) Grapes d) Barley

Q.4 Lime-stone is predominantly (CO3)

a) Magnesium carbonate

b) Sodium carbonate

- c) Calcium carbonate
- d) Potassium carbonate

Q.5 Which of the following is known as mixed fertilizer?
(CO4)

- a) Urea
- b) Ammonia nitrate
- c) Triple super phosphate
- d) NPK

Q.6 Kraft process is associated with which industry.
(CO2)

- a) Cement industry
- b) Pulp and paper industry
- c) Fertilizer Industry
- d) Sulphuric acid production

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory.
(6x1=6)

Q.7 Draw the symbol of pump used in process flowsheet.
(CO1)

Q.8 Define fermentation.
(CO3)

Q.9 Which process is used for manufacture of vegetable ghee?
(CO2)

Q.10 Name the catalyst used in the production of urea from CO_2 and NH_3 .
(CO4)

(2)

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Q.11 Define detergents.
(CO2)

Q.12 Name any two different types of cement.
(CO3)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions.
(8x4=32)

Q.13 Explain the manufacturing process of sugar from sugarcane with the help of flowsheet.
(CO3)

Q.14 Discuss the dry process for manufacturing of Portland cement.
(CO3)

Q.15 Differentiate between mechanical pulping and chemical pulping.
(CO2)

Q.16 Briefly explain different additives and their role in soaps and detergents.

Q.17 Explain the manufacturing process of urea with the help of flowsheet.
(CO4)

Q.18 Discuss any four major engineering problems associated with sulfuric acid manufacturing.
(CO4)

Q.19 Draw the schematic representation of any four equipments used in chemical process industry.
(CO1)

Q.20 Describe briefly :-
(CO1)

- a) Block diagram process flow sheets.
- b) Utility line diagram.

(3)

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